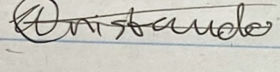
2AC Assignment CRSALEO10

Honesty Declaration:

I am aware of the consequences of engaging in Academic Dishonesty. This work is the result of my sole contribution



(a) We want $\lim_{(x,y)\to(0,0)} f_1(x,y)$

Now $f_1(x,y) = \frac{x^2\sin x - y\sin y + y\sin x - x^2\sin y}{y+x^2}$

$= \frac{x^2(\sin x - \sin y) + y(\sin x - \sin y)}{x^2+y}$

$= \frac{(x^2+y)(\sin x - \sin y)}{x^2+y}$

$= \sin x - \sin y \quad \forall x,y \in \mathbb{R},\ x^2+y \neq 0 \Rightarrow x \neq 0$ and $y \neq 0$

So $\lim_{(x,y)\to(0,0)} f_1(x,y)$ the limit exists and $= \lim_{(x,y)\to(0,0)} \sin x - \sin y = \sin 0 - \sin 0 = 0$

(b) We want $\lim_{(x,y)\to(0,0)} f_2(x,y)$

Now $f_2(0,y) = \frac{0\ln(1+|y|)}{\sqrt{|x|^3+y^2}} = 0$, $\forall x,y \in \mathbb{R}$, $x \neq 0$ and $y \neq 0$ ($\sqrt{|x|^3+y^2} = 0 \Rightarrow$)

Also $f_2(x,0) = \frac{x\ln(1)}{\sqrt{|x|^3+y^2}} = 0 \quad \forall x,y \in \mathbb{R}, x \neq 0$ and $y \neq 0$

So $\lim_{y\to 0} f_2(0,y) = 0 = \lim_{x\to 0} f(x,0)$ meaning the limit is 0 along the x and y axis.

So take $L=0$ to be the candidate solution:

Then $0 \leq \left|\frac{x\ln(1+|y|)}{\sqrt{|x|^3+y^2}} - 0\right| = \frac{|x\ln(1+|y|)|}{\sqrt{|x|^3+y^2}}$ since $\sqrt{|x|^3+y^2} \geq 0$

Now note that $|y| \geq 0 \Rightarrow 1+|y| \geq 1 \Rightarrow \ln(1+|y|) \geq 0$, so:

$\frac{|x\ln(1+|y|)|}{\sqrt{|x|^3+y^2}} = \frac{|x||\ln(1+|y|)|}{\sqrt{|x|^3+y^2}} = \frac{|x|\ln(1+|y|)}{\sqrt{|x|^3+y^2}}$

Since $|y| \geq 0$, we use the inequality $\ln(1+t) \leq t, \forall t \geq 0$

$\Rightarrow \ln(1+|y|) \leq |y|$

So $\frac{|x|\ln(1+|y|)}{\sqrt{|x|^3+y^2}} \leq \frac{|x||y|}{\sqrt{|x|^3+y^2}} \leq \frac{|x||y|}{\sqrt{y^2}}$ since $|x|^3 \geq 0$

But $\frac{|x||y|}{\sqrt{y^2}} = \frac{|x||y|}{|y|} = |x|$ and $\lim_{x\to 0}|x| = 0$, so the limit exists by the Squeeze Theorem, and $\lim_{(x,y)\to(0,0)} f_2(x,y) = 0$

c) First note that $f_3(x,0) = \frac{0}{x^2}\left(\sin\left(\frac{x}{\sqrt{x^2}}\right)\right) = 0 \quad \forall x \in \mathbb{R} - \{0\}$

$\therefore \lim_{x\to 0} f_3(x,0) = 0$ so along x-axis, limit is 0

Also $f_3(x,x) = \frac{x^2}{x^2}\sin\left(\frac{x}{\sqrt{x^2+x^2}}\right) = \sin\left(\frac{x}{\sqrt{2}|x|}\right) \quad \forall x \neq 0$

But $\sin\left(\frac{x}{\sqrt{2}|x|}\right) = \begin{cases} \sin\left(\frac{x}{\sqrt{2}x}\right) & x > 0 \\ \sin\left(-\frac{x}{\sqrt{2}x}\right) & x < 0 \end{cases} = \begin{cases} \sin\left(\frac{1}{\sqrt{2}}\right) \\ \sin\left(-\frac{1}{\sqrt{2}}\right) \end{cases}$

But then we have $\lim_{x\to 0^+} f_3(x,x) = \sin\left(\frac{1}{\sqrt{2}}\right) \neq 0$

so $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist